

LAB #3: GAZEBO AND RVIZ2

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Abstract — This lab demonstrates how to create a basic ROS 2 package that integrates Gazebo simulation, RViz visualization, and launch files in ROS 2 Humble. For that we need the following prerequisites:

- ROS 2 Humble installed
- Gazebo Garden (or Gazebo Classic)
- RViz2

A. Create a ROS 2 Package

First, create a new ROS 2 workspace and package:

```
# Create workspace
mkdir -p ~/ros2_ws/src
cd ~/ros2_ws/src

# Create package
ros2 pkg create --build-type ament_python
basic_gazebo_demo --dependencies rclpy
geometry_msgs sensor_msgs nav_msgs tf2_ros
gazebo_ros_pkgs

cd basic_gazebo_demo
```

B. Package Structure

Your package should have the following structure:

```
basic_gazebo_demo/
├── package.xml
├── setup.py
├── setup.cfg
├── resource/
│   └── basic_gazebo_demo
├── basic_gazebo_demo/
│   └── __init__.py
├── launch/
│   └── gazebo_rviz.launch.py
├── worlds/
│   └── simple_world.world
├── models/
│   └── simple_robot/
│       ├── model.sdf
│       └── model.config
├── rviz/
│   └── robot_config.rviz
```

C. Package Configuration

Update package.xml:

```
<?xml version="1.0"?>
<?xml-model href="http://download.ros.org/
schema/package_format3.xsd"
schematypens="http://www.w3.org/2001/XMLSchema"?
>
<package format="3">
  <name>basic_gazebo_demo</name>
  <version>0.0.0</version>
  <description>Basic Gazebo, RViz, and Launch
File Demo</description>
  <maintainer
email="student@example.com">Student</maintainer>
  <license>Apache-2.0</license>

  <buildtool_depend>ament_python</
buildtool_depend>

  <depend>rclpy</depend>
  <depend>geometry_msgs</depend>
  <depend>sensor_msgs</depend>
  <depend>nav_msgs</depend>
  <depend>tf2_ros</depend>
  <depend>gazebo_ros_pkgs</depend>

  <test_depend>ament_copyright</test_depend>
  <test_depend>ament_flake8</test_depend>
  <test_depend>ament_pep257</test_depend>
  <test_depend>python3-pytest</test_depend>

  <export>
    <build_type>ament_python</build_type>
  </export>
</package>
```

D. Gazebo World File

Create worlds/simple_world.world:

```
<?xml version="1.0" ?>
<sdf version="1.6">
  <world name="simple_world">
    <include>
      <uri>model://ground_plane</uri>
    </include>

    <include>
```

```

    <uri>model://sun</uri>
  </include>

  <light name="point_light" type="point">
    <pose>0 0 2 0 0 0</pose>
    <diffuse>1 1 1</diffuse>
    <specular>1 1 1</specular>
    <attenuation>
      <range>10</range>
      <constant>0.2</constant>
      <linear>0.1</linear>
      <quadratic>0.01</quadratic>
    </attenuation>
  </light>
</world>
</sdf>

```

E. Simple Robot Model

Create models/simple_robot/model.sdf:

```

<?xml version="1.0"?>
<sdf version="1.6">
  <model name="simple_robot">
    <pose>0 0 0.1 0 0 0</pose>

    <link name="base_link">
      <visual>
        <geometry>
          <box>
            <size>0.3 0.3 0.1</size>
          </box>
        </geometry>
        <material>
          <ambient>0.8 0.2 0.2 1</ambient>
          <diffuse>0.8 0.2 0.2 1</diffuse>
        </material>
      </visual>

      <collision>
        <geometry>
          <box>
            <size>0.3 0.3 0.1</size>
          </box>
        </geometry>
      </collision>

      <inertial>
        <mass>1.0</mass>
        <inertia>
          <ixx>0.1</ixx>
          <iyy>0.1</iyy>
          <izz>0.1</izz>
        </inertia>
      </inertial>
    </link>

    <joint name="left_wheel_joint"
type="revolute">

```

```

      <parent>base_link</parent>
      <child>left_wheel</child>
      <axis>
        <xyz>0 1 0</xyz>
      </axis>
    </joint>

    <link name="left_wheel">
      <visual>
        <geometry>
          <cylinder>
            <radius>0.05</radius>
            <length>0.02</length>
          </cylinder>
        </geometry>
        <material>
          <ambient>0.2 0.2 0.2 1</ambient>
        </material>
      </visual>
      <collision>
        <geometry>
          <cylinder>
            <radius>0.05</radius>
            <length>0.02</length>
          </cylinder>
        </geometry>
      </collision>
      <inertial>
        <mass>0.1</mass>
        <inertia>
          <ixx>0.001</ixx>
          <iyy>0.001</iyy>
          <izz>0.001</izz>
        </inertia>
      </inertial>
    </link>

    <joint name="right_wheel_joint"
type="revolute">
      <parent>base_link</parent>
      <child>right_wheel</child>
      <axis>
        <xyz>0 1 0</xyz>
      </axis>
    </joint>

    <link name="right_wheel">
      <visual>
        <geometry>
          <cylinder>
            <radius>0.05</radius>
            <length>0.02</length>
          </cylinder>
        </geometry>
        <material>
          <ambient>0.2 0.2 0.2 1</ambient>
        </material>
      </visual>

```

```

    <collision>
      <geometry>
        <cylinder>
          <radius>0.05</radius>
          <length>0.02</length>
        </cylinder>
      </geometry>
    </collision>
    <inertial>
      <mass>0.1</mass>
      <inertia>
        <ixx>0.001</ixx>
        <iyy>0.001</iyy>
        <izz>0.001</izz>
      </inertia>
    </inertial>
  </link>

  <plugin name="differential_drive_controller"
filename="libgazebo_ros_diff_drive.so">
    <ros>
      <namespace>/robot</namespace>
    </ros>
    <left_joint>left_wheel_joint</left_joint>
    <right_joint>right_wheel_joint</
right_joint>
    <wheel_separation>0.3</wheel_separation>
    <wheel_diameter>0.1</wheel_diameter>
    <wheel_acceleration>1.0</
wheel_acceleration>
    <wheel_torque>20</wheel_torque>
    <command_topic>cmd_vel</command_topic>
    <odometry_topic>odom</odometry_topic>
    <odometry_frame>odom</odometry_frame>
    <robot_base_frame>base_link</
robot_base_frame>
  </plugin>
</model>
</sdf>

```

Create models/simple_robot/model.config:

```

<?xml version="1.0"?>
<model>
  <name>simple_robot</name>
  <version>1.0</version>
  <sdf version="1.6">model.sdf</sdf>
  <description>A simple differential drive
robot</description>
</model>

```

F. Launch File

Create launch/gazebo_rviz.launch.py:

```

import os
from ament_index_python.packages import
get_package_share_directory
from launch import LaunchDescription

```

```

from launch.actions import
DeclareLaunchArgument, IncludeLaunchDescription
from launch.conditions import IfCondition
from launch.launch_description_sources import
PythonLaunchDescriptionSource
from launch.substitutions import
LaunchConfiguration, PathJoinSubstitution
from launch_ros.actions import Node
from launch_ros.substitutions import
FindPackageShare

def generate_launch_description():
    # Package directories
    pkg_share =
FindPackageShare(package='basic_gazebo_demo').find('basic_g

    # Launch configuration variables
    use_sim_time =
LaunchConfiguration('use_sim_time')
    world_file =
LaunchConfiguration('world_file')
    rviz_config_file =
LaunchConfiguration('rviz_config_file')

    # Declare launch arguments
    declare_use_sim_time_cmd =
DeclareLaunchArgument(
        'use_sim_time',
        default_value='true',
        description='Use simulation (Gazebo)
clock if true'
    )

    declare_world_file_cmd =
DeclareLaunchArgument(
        'world_file',
        default_value=os.path.join(pkg_share,
'worlds', 'simple_world.world'),
        description='Full path to world file to
load'
    )

    declare_rviz_config_file_cmd =
DeclareLaunchArgument(
        'rviz_config_file',
        default_value=os.path.join(pkg_share,
'rviz', 'robot_config.rviz'),
        description='Full path to RVIZ config
file to load'
    )

    # Gazebo launch
    gazebo_launch_cmd =
IncludeLaunchDescription(
        PythonLaunchDescriptionSource(
os.path.join(get_package_share_directory('gazebo_ros'),
'launch', 'gazebo.launch.py')

```

```

    ),
    launch_arguments={'world':
world_file}.items()
)

# Spawn robot in Gazebo
spawn_entity_cmd = Node(
    package='gazebo_ros',
    executable='spawn_entity.py',
    arguments=['-entity', 'simple_robot', '-
file',
                os.path.join(pkg_share,
'models', 'simple_robot', 'model.sdf')],
    output='screen'
)

# RViz launch
rviz_cmd = Node(
    package='rviz2',
    executable='rviz2',
    name='rviz2',
    arguments=['-d', rviz_config_file],
    parameters=[{'use_sim_time':
use_sim_time}],
    output='screen'
)

# Create the launch description and populate
ld = LaunchDescription()

# Declare the launch options
ld.add_action(declare_use_sim_time_cmd)
ld.add_action(declare_world_file_cmd)
ld.add_action(declare_rviz_config_file_cmd)

# Add the commands to the launch description
ld.add_action(gazebo_launch_cmd)
ld.add_action(spawn_entity_cmd)
ld.add_action(rviz_cmd)

return ld

```

```
angular: {x: 0.0, y: 0.0, z: 0.5}"
```

```

# Terminal 3: View topics
ros2 topic list
ros2 topic echo /robot/odom

```

G. Build and Run

Build the package:

```

cd ~/ros2_ws
colcon build --packages-select basic_gazebo_demo
source install/setup.bash

```

Run the complete example:

```

# Terminal 1: Launch everything
ros2 launch basic_gazebo_demo
gazebo_rviz.launch.py

# Terminal 2: Control the robot
ros2 topic pub /robot/cmd_vel geometry_msgs/msg/
Twist "linear: {x: 0.5, y: 0.0, z: 0.0}

```